

Project Executable Files

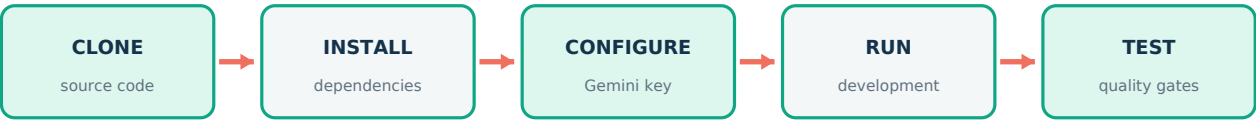
Implementation and run guide

Date	30 September 2026
Team ID	SWTID-2026-9380
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	4 Marks

Expected deliverable files

File	Purpose
README.md	Project overview, prerequisites, setup, run and test commands.
.env.example	Environment variable names only; never real keys.
package/dependency manifest	Pinned application and testing dependencies.
application source	UI, server endpoints, domain types and rendering.
prompt template	Versioned Gemini system instruction and output schema.
test suite	Unit, end-to-end, load and responsible-AI cases.
deployment configuration	HTTPS deployment and environment setup.
docs/	Architecture, data flow, test evidence and demonstration material.

Setup sequence



- Create a local environment file from the example and insert the Gemini API key only on the server.
- Install the dependencies using the repository’s declared package manager.
- Run automated tests before starting the application.
- Open the local HTTPS/development URL, complete the profile and generate a plan.
- For deployment, store the key in the hosting platform’s encrypted secret settings.

Executable-file limitation

The supplied archive is a documentation template and contains no FitBuddy source code. This report defines the expected executable package; code, screenshots and measured test evidence must be added after implementation.

Complete Project Documentation

FitBuddy product and technical summary

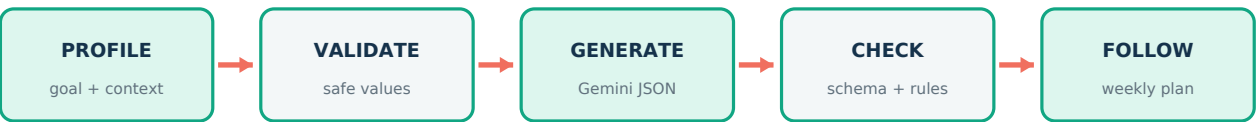
Project description

FitBuddy is an AI-assisted fitness plan generator that converts a validated user profile into a structured weekly exercise plan. It uses Gemini models for flexible plan composition while application-side schemas and safety rules control the format, completeness and responsible boundaries of the output.

Objectives

- Reduce the confusion involved in selecting exercises and organizing a week.
- Personalize plans by goal, experience, schedule, session length and equipment.
- Generate clear, actionable sessions with warm-up, dosage, rest, alternatives and cool-down.
- Support gradual progression while avoiding medical diagnosis or treatment claims.
- Protect model credentials and minimize collection of personal data.

Primary workflow



Assumptions

- Users provide accurate preferences and disclosed limitations.
- FitBuddy supports general fitness planning, not medical rehabilitation.
- A current Gemini model is selected through configuration at deployment time.
- Internet access is available during generation.
- Any stored profile or plan history is protected by authenticated access.

Responsible AI controls

Risk	Control
Hallucinated exercise or dosage	Allowlisted/validated fields, conservative ranges, output rejection and human review during evaluation.
Medical overreach	Explicit system boundary, risk-pattern checks and escalation language.
Prompt injection	Treat profile text as data, delimit it, constrain output to schema and never expose secrets.
Bias / exclusion	Test varied goals, ages, equipment access and ability constraints; provide alternatives.
Over-reliance	Show assumptions and encourage users to stop with pain and seek qualified advice.
Privacy	Minimize inputs, avoid logging profile text and provide deletion for saved history.

Project Workflow & Milestones

From model selection to release

Milestone	Activities	Deliverable
1. Discovery	Confirm personas, requirements, scope and success measures.	Approved problem and requirement set.
2. Architecture	Choose Gemini integration, schema, module boundaries and safety rules.	Architecture and prompt contract.
3. Core build	Implement profile, validation, server generation and result rendering.	End-to-end plan flow.
4. Adaptation	Add alternatives, progression, regeneration and optional history.	Personalized plan versions.
5. Quality	Run functional, accessibility, security, load and responsible-AI tests.	Signed test report with evidence.
6. Release	Configure secrets, deploy, monitor and prepare rollback.	Working release and runbook.
7. Demonstration	Rehearse scenarios, assign speakers and collect feedback.	Recorded/live project demonstration.

AI model selection criteria

- Structured-output reliability and instruction following.
- Latency and cost under expected usage.
- Model availability in the deployment region and account.
- Safety behavior across the project's evaluation set.
- Ability to change the configured model without rewriting business logic.

Constraints and limitations

- Generated plans can contain mistakes and require validation.
- Exercise technique cannot be assessed without video or sensor input.
- A user's medical readiness cannot be established by a short form.
- Provider latency, quotas and model changes can affect generation.
- Progress recommendations are only as reliable as user-entered feedback.

API & Data Contract

Implementation-ready interface specification

Operation	Input	Output	Errors
Generate plan	Validated profile object	Validated Plan object + version	400 invalid profile; 422 unsafe/unusable output; 429 capacity; 503 provider unavailable
Regenerate plan	Profile + prior plan version + changed preferences	New Plan object + version	Same as generation; never overwrite silently
Submit check-in	Plan version, completed sessions, difficulty, notes	Acknowledgement + adjustment summary	400 invalid range; 404 missing plan
Delete history	Authenticated user request	Deletion confirmation	401 unauthenticated; 404 no history

Minimum profile fields

Field	Type / allowed values	Required
goal	strength, endurance, mobility, general fitness, weight management	Yes
experienceLevel	beginner, intermediate, advanced	Yes
daysPerWeek	integer 1-6	Yes
minutesPerSession	integer 15-120	Yes
equipment	controlled list + "none"	Yes
preferences	short free text, length-limited	No
limitations	short free text with safety notice	No
progressFeedback	completion and perceived difficulty 1-5	No

Privacy rule

Do not require a full name, exact address, government identifier or unrelated health data to generate a general fitness plan.